# Gears, motor and first power



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Compare a gear-based prediction with a supervised first powered test.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 8, Day 3; 3 course days later.*
Ideal speed becomes 3 times as large. Actual rpm, load or slipping may change.

**02. Explain a worked example (worked)**
One rotation 0.20 m; 2 rotations/s gives 0.40 m/s.

**03. Look and explain (model)**
Actual labeled configuration with inspection observations.

**04. Build the solution (practice)**
.30 m/s.

**05. Use the idea (reason)**
Loaded rpm, slip, gear rubbing, wheel circumference; measure loaded wheel rpm or marked-wheel rotations over distance.

**06. Choose a method (mixed)** *Review: Unit 7, Day 12; 12 course days later.*
A stronger field at A in that drawing convention; field lines are a model, not physical strands.

**07. Exit check (exit)**
Actual checkpoint and evidence; leave approval unmarked until granted.

**08. STRETCH: Apply it another way (stretch)**
Slip, wrong effective circumference, timing/rotation measurement or changing speed. Track wheel and ground motion together.

**Teaching note:** Do not record a prediction as a measured rpm. Label no-load and loaded measurements separately.

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**Learning target:** Compare a gear-based prediction with a supervised first powered test.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 8, Day 3; 3 course days later.*
Torque ratio 5; speed ratio 1/5. Assumes ideal gears with negligible losses.

**02. Explain a worked example (worked)**
One rotation 0.20 m; 2 rotations/s gives 0.40 m/s.

**03. Represent and interpret (model)**
Actual labeled configuration with inspection observations.

**04. Solve and justify (practice)**
160 wheel rpm; .4189 m/s; 7.16 s.

**05. Analyze the reasoning (reason)**
Loaded rpm, slip, gear rubbing, wheel circumference; measure loaded wheel rpm or marked-wheel rotations over distance.

**06. Choose a method (mixed)** *Review: Unit 7, Day 12; 12 course days later.*
Each piece is normally a smaller magnet with both poles; cutting does not isolate a magnetic monopole.

**07. Exit check (exit)**
25% reduction; drivetrain load lowers actual rotational speed.

**08. STRETCH: Apply it another way (stretch)**
Ideal $2\pi \times 0.025 \times 4 \approx 0.628$ m/s; ratio$\approx 0.796$. Other measurement/model errors also possible.

**Teaching note:** Do not record a prediction as a measured rpm. Label no-load and loaded measurements separately.